

# DUAL-ARM COLLABORATIVE MANIPULATION

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Kobuki-Based Mobile Bimanual Manipulator

Gazebo Harmonic Simulation | ROS 2 Jazzy

**Perception -> Approach -> Align -> IK Grasp -> Co-Manipulation -> Reorientation -> Placement**

An autonomous mobile manipulator that surveys a room, identifies everyday objects lying in an unstable orientation, drives to them, grasps them with two cooperating 4-DOF arms, rotates them upright in mid-air, and places them standing on a shelf - repeating the cycle for every fallen object it has detected.

## **Simulation stack**

ROS 2 Jazzy, Gazebo Harmonic (gz-sim 8), gz\_ros2\_control, Pop!\_OS 24.04

## **Workspace**

bimanual\_ws - 3 ROS 2 packages, 4 simulation worlds, 2 autonomous nodes

## **Document scope**

Design, implementation, runtime workflow, verification status and open issues

# 1. Objective and Task Definition

What the system is required to do, and the seven phases it is decomposed into

## Real-world task

Locate a daily object that has fallen over somewhere in a room, retrieve it with two arms working together, correct its orientation in mid-air so that its stable base faces downwards, and set it upright on a designated shelf. The value of the bimanual arrangement is precisely the mid-air reorientation: a single 4-DOF arm cannot rotate a held object about a horizontal axis without releasing it.

## Phase decomposition implemented in simulation

| # | Phase           | What happens                                                               | Responsible component                        |
|---|-----------------|----------------------------------------------------------------------------|----------------------------------------------|
| 1 | SEARCH          | Full survey rotation; every object colour seen is recorded                 | mission_node.rotate_by + perception_node     |
| 2 | APPROACH        | Visual servo toward the target; blind odometry creep for the final stretch | mission_node.approach                        |
| 3 | ALIGN           | Bearing driven to zero before the creep so the object axis stays aligned   | mission_node.center_on                       |
| 4 | IK GRASP        | Both grippers descend from above onto the lying object and close           | choreography.grasp_waypoints + ik.solve      |
| 5 | CO-MANIPULATION | A rigid constraint binds the object to the gripper; both arms move as one  | DetachableJoint + mission_node.set_grasp     |
| 6 | REORIENTATION   | Two-phase lift then 90 deg pitch flip to upright                           | choreography.flip_waypoints                  |
| 7 | PLACEMENT       | LiDAR-docked approach to the shelf, lower, release, retract, back away     | mission_node.dock_to_shelf + place_waypoints |

### Scope of autonomy

One launch command runs the entire cycle for three different objects in sequence with no operator input: detection, navigation, grasping, reorientation, placement and recovery from per-object failures are all closed-loop.

## 2. Platform Geometry - as Built from the Design Blueprint

Every dimension below is a single named constant in robot.urdf.xacro, reused by the kinematics module

### Base and deck

|                   |                                                  |
|-------------------|--------------------------------------------------|
| Base radius       | 0.1775 m (Kobuki, 354 mm dia.)                   |
| Base body height  | 0.125 m                                          |
| Arm mounting deck | 0.140 m AGL (base + 15 mm standoff)              |
| Drive wheels      | $r = 0.038$ m, separation 0.230 m, diff-drive    |
| Casters           | front/rear spheres at $\pm 0.15$ m, frictionless |

### Arm pair (4-DOF each)

|                     |                                                                           |
|---------------------|---------------------------------------------------------------------------|
| Mount positions     | $x = +0.095$ m, $y = \pm 0.08875$ m ( $r = 0.130$ m, separation 0.1775 m) |
| Shoulder pitch axis | 0.180 m AGL (deck + 40 mm riser)                                          |
| Link lengths        | $L1 = L2 = 0.160$ m; grasp point 0.070 m beyond the wrist                 |
| Geometric max reach | 0.320 m; practical band 0.20-0.25 m                                       |
| Joint limits        | yaw $\pm 150$ deg, three pitches $\pm 115$ deg                            |
| Gripper             | two prismatic fingers, 75 mm long, 8-65 mm travel, high-friction pads     |

### Sensor mast - twin rods per the blueprint top view

|            |                                                                                                                         |
|------------|-------------------------------------------------------------------------------------------------------------------------|
| Rods       | two vertical posts, each 0.14 m from base centre on the rear half, symmetric about the centreline, joined by a crossbar |
| 2D LiDAR   | 0.46 m AGL, 360 samples, 0.12-8.0 m                                                                                     |
| RGB camera | 0.42 m AGL, pitched 15 deg down, 640x480, 69 deg HFOV                                                                   |

### Payload and object envelope

Payload target  $\leq 500$  g against a Kobuki-class capacity of  $\sim 5$  kg. Objects handled: 60 mm dia. x 200 mm bottle (150 g), 80 mm dia. x 100 mm cup-bowl (200 g), 60 x 80 x 180 mm box (250 g) - all inside the 5-18 cm diameter, 5-25 cm height envelope of the design sheet.

### 3. Software Architecture

Three ROS 2 packages with a strict separation between description, simulation assets and autonomy

| Package                     | Build type   | Contents                                                                                                                | Role                                                                                                                     |
|-----------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| kobuki_bimanual_description | ament_cmake  | robot.urdf.xacro, base.xacro, mast_sensors.xacro, arm.xacro, ros2_control.xacro, gazebo_plugins.xacro, controllers.yaml | Single source of truth for geometry, sensors, hardware interfaces and controller configuration                           |
| kobuki_bimanual_gazebo      | ament_cmake  | 4 worlds (main + one per object), 4 models (bottle, bowl, cracker_box, shelf), sim.launch.py                            | Simulation environment and the single bring-up launch file that starts and orders every process                          |
| kobuki_bimanual_behavior    | ament_python | ik.py, choreography.py, arm_commander.py, perception_node.py, mission_node.py, mission_params.yaml, test_ik.py          | All autonomy: perception, analytic kinematics, bimanual trajectory generation, mission state machine, offline test suite |

#### Design principles applied

- Geometry is declared once. The kinematics module mirrors the URDF constants and an offline test asserts that every motion the mission can command is reachable within the declared joint limits.
- Autonomy is pure Python and simulator-independent: kinematics and choreography carry no ROS or Gazebo imports, so they are unit testable without starting a simulation.
- All tunable numbers - HSV ranges, grasp offsets, stand-off distances, placement positions, speed caps - live in one parameter file, not in code.

## 4. Runtime Process and Node Map

Which process is responsible for what, and the interfaces between them

| Process / node                  | Responsibility                                                                                                           | Consumes                                              | Produces                                                      |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------|
| gz sim (server + GUI)           | Rigid-body physics, contact, gravity, sensor rendering; hosts the DiffDrive, DetachableJoint and gz_ros2_control plugins | joint commands, wheel twist                           | sensor data, poses, clock                                     |
| robot_state_publisher           | Expands the xacro and publishes the robot model and kinematic TF tree                                                    | /robot_description                                    | /tf, /tf_static                                               |
| ros_gz_sim create               | Spawns the robot into the running world, then exits                                                                      | /robot_description                                    | spawn request                                                 |
| ros_gz_bridge                   | Translates every Gazebo transport topic to and from ROS 2                                                                | gz topics                                             | /clock /scan /camera /camera_info /odom /tf, grasp state      |
| controller_manager (in-process) | Loads and activates four controllers over 12 joint interfaces                                                            | controllers.yaml                                      | /joint_states, joint position commands                        |
| perception_node                 | Colour segmentation, ground-plane ranging, lying-vs-upright estimation                                                   | /camera, /camera_info                                 | /detections (JSON)                                            |
| mission_node                    | The mission state machine: survey, approach, grasp, reorient, dock, place, recover                                       | /detections /odom /joint_states /scan /grasp/* /state | /cmd_vel, arm trajectories, gripper commands, attach & detach |

### Controllers

|                                                   |                                                                    |
|---------------------------------------------------|--------------------------------------------------------------------|
| <b>joint_state_broadcaster</b>                    | publishes all 12 joint states at 200 Hz                            |
| <b>left_arm_controller / right_arm_controller</b> | JointTrajectoryController, spline interpolation, 4 joints each     |
| <b>gripper_controller</b>                         | JointGroupPositionController over the four prismatic finger joints |

## 5. Simulation Workflow (1 of 2) - Bring-up and Target Acquisition

Deterministic ordering is enforced; nothing begins before the previous stage is confirmed

| Stage                 | What the robot does                                                                                          | Code path                                               |
|-----------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Launch                | World loads, robot is spawned, bridges open                                                                  | sim.launch.py: gz_sim include, create, parameter_bridge |
| Controller bring-up   | Four controllers are activated strictly one after another                                                    | sim.launch.py: chain of four OnProcessExit handlers     |
| Readiness gate        | Mission waits until all 12 joints report state                                                               | mission_node.run: wait on ARM_JOINTS                    |
| Startup safety        | All grasp constraints released with confirmation, grippers opened, arms tucked                               | set_grasp(..., False), grippers(), tuck()               |
| SEARCH                | One full survey rotation (360 deg + 17 deg overlap) recording every colour seen                              | rotate_by( $2\pi$ + 0.3, note_seen=...)                 |
| Per object: re-centre | Robot returns to a central scan hub so every approach direction is repeatable                                | goto(scan_hub)                                          |
| Reacquire             | Rotates until four consecutive detections of that specific colour; a second vantage point is tried if needed | rotate_until_found, then goto(0.45, 0)                  |
| Triage                | Objects already standing are skipped using the estimated physical height                                     | est_h >= 0.7 * upright_h                                |
| APPROACH              | Visual servo until 0.55 m, precise re-centre, then blind odometry creep to the 0.30 m grasp stand-off        | approach(): centre_on(tol 0.015) then drive_odom        |
| Safety                | Forward LiDAR guard aborts an approach that would collide                                                    | front_min < front_guard (0.35 m)                        |

## 5. Simulation Workflow (2 of 2) - Manipulation and Placement

The bimanual sequence, from descent to a released object standing on the shelf

| Stage            | What the robot does                                                                                          | Code path                                                    |
|------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Staging          | Arms move to a raised pose so interpolation cannot sweep through the base deck                               | choreography.STAGE_LEFT / STAGE_RIGHT                        |
| Staggered deploy | Left hand reaches its hover station first, then the right, so the arms never cross                           | ik.solve per side, published separately                      |
| IK GRASP         | Both grippers descend vertically onto the lying object and close to a light-touch fit                        | grasp_waypoints -> pair_trajs -> ik.solve -> JointTrajectory |
| Constraint       | A rigid joint binds the object to the palm; success is confirmed on the plugin state topic before proceeding | set_grasp(name, True) with retries                           |
| Finger relief    | Fingers back off 12 mm: the constraint carries the load, so finger contact must not fight it                 | grippers(close + 0.012)                                      |
| REORIENTATION    | Phase A lifts and pre-rotates; phase B rotates to 90 deg about the object centre at a fixed hover point      | flip_waypoints: 4 lift + 8 rotation waypoints                |
| Transit          | Loaded drive to the shelf in a speed- and turn-rate-limited mode                                             | goto(shelf_approach, slow=True)                              |
| Docking          | The shelf back-board is found in the LiDAR scan and the robot servos to a measured 0.594 m stand-off         | dock_to_shelf -> _dock_servo -> _board_cluster               |
| PLACEMENT        | Object lowered to its own shelf height at its own lateral slot, then released with confirmation              | place_waypoints(y_off) then set_grasp(name, False)           |
| Withdraw         | Vertical clearance, reverse 0.6 m, arms tucked, next object                                                  | clear_waypoints, drive_odom(-0.6), tuck()                    |
| Fault isolation  | Any failure releases the object, recovers the pose and continues with the next object                        | per-object try / except in run()                             |

## 6. Perception Design

Why the camera - not the LiDAR - finds the objects, and how range is obtained from a single RGB image

### Sensor division of labour

A horizontal 2D scan plane cannot intersect a 5-10 cm object lying on the floor from any practical mast height. The LiDAR is therefore used for room structure, the forward safety guard and shelf docking, while object detection and ranging are done monocularly.

### Pipeline, executed per camera frame at 15 Hz

- Convert to HSV and threshold one colour band per object class (orange bottle, blue bowl, green box); take the largest contour above a minimum pixel area.
- Range by ground-plane back-projection: the depression angle of the blob's bottom pixel row, combined with the known camera height of 0.42 m and 15 deg tilt, yields the horizontal distance to the floor contact point.
- Bearing from the horizontal pixel offset and the focal length (intrinsics taken from /camera\_info when available).
- Apparent physical height from the blob height and that range - a lying bottle measures ~6 cm, an upright one ~20 cm, which is the lying-vs-upright test.
- Results are published as one JSON message on /detections, so the mission consumes a single self-describing topic.

#### Practical lesson recorded during development

Colour segmentation is only as good as the scene palette. The wooden shelf shared a hue family with the orange bottle, and the original blue arm links fell inside the bowl mask - both produced confident false detections. The shelf and the arms were desaturated and saturation floors were raised, which removed the ambiguity by construction rather than by tuning thresholds.

## 7. Kinematics, Control and the Grasp Constraint

Analytic inverse kinematics, synchronised dual-arm trajectories, and the modelling choice for the grasp

### Analytic 4-DOF inverse kinematics (ik.py)

- Yaw about the vertical axis through the shoulder decouples exactly, reducing the problem to a planar three-pitch chain; the elbow follows from the cosine rule and the wrist absorbs the remaining orientation error.
- Both elbow-up and elbow-down solutions are generated; the elbow-up branch is preferred and any solution violating a joint limit is rejected, raising a typed error the mission can handle.
- Forward kinematics is provided as well, which lets the test suite assert round-trip consistency to  $1e-9$  m over the entire workspace.

### Trajectory generation and execution

- Cartesian waypoint pairs are solved per arm and emitted as JointTrajectory messages with matched timestamps, so the two arms stay synchronised through the flip.
- The joint trajectory controllers interpolate with splines; the position-tracking gain of the simulated hardware interface was raised from the default 0.1 to 15.0 after the default was found to produce roughly a ten-second joint lag.

### The grasp constraint - an explicit modelling decision

As stated in the design blueprint, the two-arm rigid grasp is realised as a fixed joint between palm and object rather than by friction contact, to avoid contact-friction instability during co-manipulation. The joint is created and destroyed at run time and its state is confirmed on a feedback topic before the mission proceeds. This is a deliberate simplification and is the main sim-to-real gap in the current model.

## 8. Closed-Loop Shelf Docking

Why odometry alone is not trusted for the placement pose, and how the shelf is located instead

### The problem

Arm reaction torques transmitted through a rigidly held payload slip the wheels during the flip and the loaded turn. Because the choreography is identical every cycle, the resulting odometry error is repeatable rather than random - which means a hard-coded odometric shelf pose fails the same way every time.

### The mechanism now implemented

- The shelf carries a 0.40 m wide, 0.60 m tall back-board specifically so that it intersects the 0.46 m scan plane and is detectable.
- Scan returns are clustered; a candidate must match the board's physical width (0.15-0.70 m as computed from its angular span and range) and must stand out in depth from its surroundings, which rejects wall fragments.
- The search sweep is bounded to  $\pm 40$  deg about the arrival heading, so a target behind the robot can never be acquired.
- The robot then servos in range and bearing to a measured 0.594 m stand-off, i.e. the placement pose is defined by the sensed shelf, not by dead reckoning.
- A successful dock is cross-checked against odometry and rejected as a phantom lock if the two disagree by more than 0.6 m; failures withdraw 0.35 m, re-approach and retry before any placement is attempted.

### Measured outcome

In the most recent run this mechanism docked the cup-bowl at a board range of 0.586 m and a bearing of 0.035 rad against a 0.594 m target - and refused to place blindly when the board could not be seen for the first object, falling back to a bounded odometric pose instead of an arbitrary one.

## 9. Verification Methodology

What is proven offline, what is only observable in simulation, and what that separation has bought

### Offline test suite - runs without ROS or Gazebo

| Suite                     | What it asserts                                                                                                                                                      |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FK/IK round trip          | Every reachable pose on a workspace grid solves and re-projects to itself within 1e-9 m for both arms                                                                |
| Unreachable rejection     | Targets outside the workspace raise a typed error rather than returning a silent bad solution                                                                        |
| Pose safety               | Tuck and staging poses are inside joint limits and keep the hands above the deck and behind the reach limit                                                          |
| Choreography reachability | Every grasp, lift, flip, place and clearance waypoint for all three objects is reachable by both arms within limits - the full mission is validated before it is run |

### Static verification

21 XML documents (xacro, SDF, manifests) parsed, 9 Python modules compiled, and a cross-package consistency sweep that checks joint names, controller names, topic names and model names agree across the description, simulation and behaviour packages.

### Defects caught before ever reaching the simulator

- The elbow pitch limit of +/-110 deg made low grasps unreachable - widened to +/-115 deg.
- A single-arc mid-air flip drove the lower wrist into an unreachable region - redesigned as a two-phase lift-then-rotate.
- A proposed staging pose overreached to 0.384 m, and a first choice of lateral placement offsets (+/-0.13 m) exceeded the cross-reaching arm's envelope - both corrected to verified values.

#### Why this matters for the report

The choreography test is the strongest artefact in the project: it converts "the arms might not reach" from a runtime surprise into a build-time failure, and it caught three real geometric defects that would otherwise have appeared as unexplained simulator behaviour.

## 10. Current Status - What Works

Behaviour observed in the most recent end-to-end runs

- Deterministic bring-up: all four controllers activate in a fixed order and the mission refuses to start until all 12 joints report state. An earlier race in which the right arm was still inactive during a grasp is eliminated.
- Survey detection: all three objects are reliably found during a single 360 deg survey rotation and logged before any motion toward them.
- Approach: repeatable to within a few centimetres - the blind odometry creep after the camera loses sight of the object measured 0.29-0.34 m across objects and runs, as expected for the stand-off geometry.
- Bimanual grasp, rigid co-manipulation and the two-phase mid-air flip to upright complete for the bottle, the cup-bowl and the box.
- Shelf docking closes the loop on the sensed board (0.586 m measured against a 0.594 m target for the cup-bowl).
- Placement is now distributed across the shelf: each object has its own lateral slot, so a newly placed object cannot knock over one placed earlier.
- Fault isolation: a failure on one object releases its constraint, restores a safe pose, backs away and continues with the next object; the mission always terminates in a reported state rather than hanging.
- Four worlds are provided: a combined world with three fallen objects handled successively, plus one world per object selectable from the same launch file.

# 11. Open Issues and Errors (1 of 2)

Observed failures, diagnosed cause, and current state of each

| ID | Observed behaviour                                                                                                                                                          | Diagnosed cause                                                                                                                                                                                                                           | State                                                                                             |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| E1 | For the first object the shelf board is not found; docking reports "notfound" twice and the object is placed at the fallback odometric pose instead of on the shelf.        | Most probable cause is occlusion of the board by the carried object and extended arms during the loaded approach. The same code docked the cup-bowl successfully in the same run, so the detector itself is sound when the view is clear. | OPEN - hypothesis not yet confirmed; instrumented cluster logging is the next diagnostic step     |
| E2 | The held object visibly shifts and appears to pass through the gripper during the flip.                                                                                     | Finger contact fighting the rigid grasp constraint: both the fingers and the fixed joint were trying to control the same body.                                                                                                            | FIX APPLIED - fingers relieved 12 mm after the constraint is confirmed; awaiting run verification |
| E3 | An object stayed welded to the palm after placement, which then pulled a placed object off the shelf, hid the next object from the camera and snagged the arm on the shelf. | The release command was fire-and-forget; a single dropped message left the constraint silently active and every later stage inherited the fault.                                                                                          | FIXED - attach and release are now confirmed on the plugin state topic with retries               |
| E4 | Arm and carried object snagged the shelf back-board while turning at the dock, stalling the robot.                                                                          | The swept radius of the arms plus payload is about 0.45 m, larger than the 0.5 m stand-off allows for a turn in place.                                                                                                                    | FIXED - a 0.35 m withdrawal is now mandatory before any re-approach turn                          |

## 12. Open Issues and Errors (2 of 2)

Remaining items, plus the single most instructive defect of the project

| ID | Observed behaviour                                                        | Diagnosed cause                                                                                                                                  | State                                                                                                                        |
|----|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| E5 | An object recorded during the survey was later reported lost and skipped. | A single vantage point was used for reacquisition; a transient occlusion was enough to abandon the object.                                       | FIXED - a second vantage point is tried before an object is abandoned                                                        |
| E6 | Placement pose drifted by metres from the intended shelf pose.            | Wheel slip under arm reaction torque during the flip and the loaded turn; an unclamped turn command was allowing up to 4.7 rad/s while carrying. | MITIGATED - turn rate clamped, a slow carry mode added, flip made gentler; LiDAR docking exists to bypass the residual drift |
| E7 | Two objects were placed on the same spot and knocked each other over.     | Placement height was per-object but the lateral position was not - every object was released at the same point on the slab.                      | FIXED - per-object lateral slots at -0.10, 0.00 and +0.10 m                                                                  |

### The most instructive defect: a single default value

gz\_ros2\_control defaults its position tracking gain to 0.1, which gave the joints roughly a ten-second lag. The arms never reached their commanded poses, sagged under gravity, dragged on the floor while driving - which corrupted odometry - and appeared in screenshots to have collapsed. Several apparently unrelated failures (unreachable grasps, wrong placement locations, "collapsed" arms) were all downstream of that one number, now set to 15.0.

# 13. Known Design Limitations

Stated explicitly - these are modelling choices and scope boundaries, not unexplained behaviour

|                               |                                                                                                                                                                                                                                                |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Grasp model</b>            | A rigid fixed joint, as specified in the design blueprint, rather than force closure through friction. Hardware would require compliant pads, effort-based grip detection and a coordinated impedance controller.                              |
| <b>Motion planning</b>        | Analytic inverse kinematics with hand-designed choreography; no MoveIt 2, therefore no general collision-aware planning. Base-deck and arm-crossing collisions are avoided by staging poses and staggered deployment rather than by a planner. |
| <b>Navigation</b>             | Odometry plus LiDAR docking; no SLAM, no global map, no Nav2 obstacle avoidance. Adequate for an open floor, not for a cluttered room.                                                                                                         |
| <b>Object pose assumption</b> | Objects are assumed to lie with their long axis pointing toward the approach direction, which the central scan hub enforces. Arbitrary in-plane orientation would need point-cloud pose estimation.                                            |
| <b>Perception modality</b>    | Single-colour segmentation per object class; sensitive to lighting and to palette collisions with the environment, as encountered during development.                                                                                          |
| <b>Orientation authority</b>  | 4-DOF per arm (yaw plus three coplanar pitches) provides planar orientation only - sufficient for the sideways-to-upright task, not for general 3D reorientation.                                                                              |
| <b>Self-collision</b>         | Disabled, as is standard practice for solver stability, so the model will not physically resist a self-intersecting command; correctness is enforced by the offline reachability and pose tests instead.                                       |

# 14. Path to an Industry-Grade System

Ordered by payoff per unit of effort

## Immediate - smoothness and robustness, small contained changes

- Emit velocities with each trajectory point so the controllers produce cubic rather than linear interpolation - the single largest visual improvement available.
- Trapezoidal velocity profiles and blended arcs for the base, replacing the current turn-drive-turn motion.
- Drive the arms through the FollowJointTrajectory action and use its result callback, eliminating the polling and settle heuristics entirely.
- Low-pass filter the perception bearing and range feeding the servo loops.

## Structural - what makes the architecture defensible

- MoveIt 2 for both arms: planning groups, real collision checking and time-parameterised trajectories, which makes an arm-through-base fault structurally impossible.
- Nav2 with SLAM or AMCL for the base, replacing dead reckoning, the scan hub heuristic and the bespoke docking routine.
- A behaviour tree in place of the hand-rolled state machine, making recovery paths explicit and introspectable.
- Depth camera and point-cloud clustering with principal-axis estimation, which removes the object-orientation assumption altogether.

## Sim-to-real prerequisites

- Force-closure grasping with effort-based verification before lift, replacing the rigid constraint.
- Safety layer: software emergency stop, command watchdog, controller-enforced workspace limits, guard active in every motion state.
- Continuous integration running the test suite and a headless mission test, plus a container image for reproducible runs.

# 15. Summary

Where the project stands

## Delivered

A complete, buildable ROS 2 Jazzy and Gazebo Harmonic simulation of a Kobuki-based dual-arm mobile manipulator, built to the dimensions of the design blueprint, that autonomously surveys a room, detects fallen objects by colour with monocular ranging, approaches them, grasps them with two cooperating 4-DOF arms, reorients them upright in mid-air, docks to a shelf using LiDAR and places each object in its own slot - repeating for every object detected and recovering per object from failures.

## Engineering position

- Geometry, kinematics and the entire motion choreography are validated offline before any simulation runs, and that suite has already caught three geometric defects.
- The failure modes that remain are contained: an unlocatable shelf now yields a bounded fallback placement rather than an arbitrary one, and a failure on one object cannot end the mission.
- One open issue is carried honestly: the first object's shelf docking failure (E1), with a stated hypothesis and a defined next diagnostic step.

## Immediate next steps

- Instrument the docking cluster rejection reasons and confirm or refute the occlusion hypothesis for E1.
- Verify in simulation the applied fixes for the grasp-constraint and finger-interference issues.
- Then proceed to the trajectory-smoothness work, which is small, contained and highly visible in a demonstration.